

OBSERVATION RECORD

Full Stack Web Development

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Week: 5

Task 1

Aim: Interactive Product Calculator: Create a webpage containing a product name, quantity, and price. Calculate total dynamically.

Source Code:

```
<!DOCTYPE html>
<html>
<head><title>Product Bill Calculator</title></head>
<body>
  <h3>Live Bill Calculator</h3>
  <p>Item: <input type="text" id="itemName"></p>
  <p>Amount: <input type="number" id="itemQty" value="1"></p>
  <p>Cost (Rs): <input type="number" id="itemCost" value="0"></p>
  <h4>Final Bill: Rs <span id="finalBill">0</span></h4>
  <script>
    // fetching inputs
    const qtyInput = document.querySelector('#itemQty');
    const costInput = document.querySelector('#itemCost');
    const billOutput = document.querySelector('#finalBill');

    // dynamically calc on input
    const updateBill = () => {
      billOutput.innerHTML = (qtyInput.value * costInput.value).toString();
    };
    qtyInput.addEventListener('input', updateBill);
    costInput.addEventListener('input', updateBill);
  </script>
</body>
</html>
```

Output:

```
C:\Users\anits-csm>cd C:\Users\anits-csm\Desktop\NodeModule

C:\Users\anits-csm\Desktop\NodeModule>node app.js
30

C:\Users\anits-csm\Desktop\NodeModule>|
```

Test Cases:

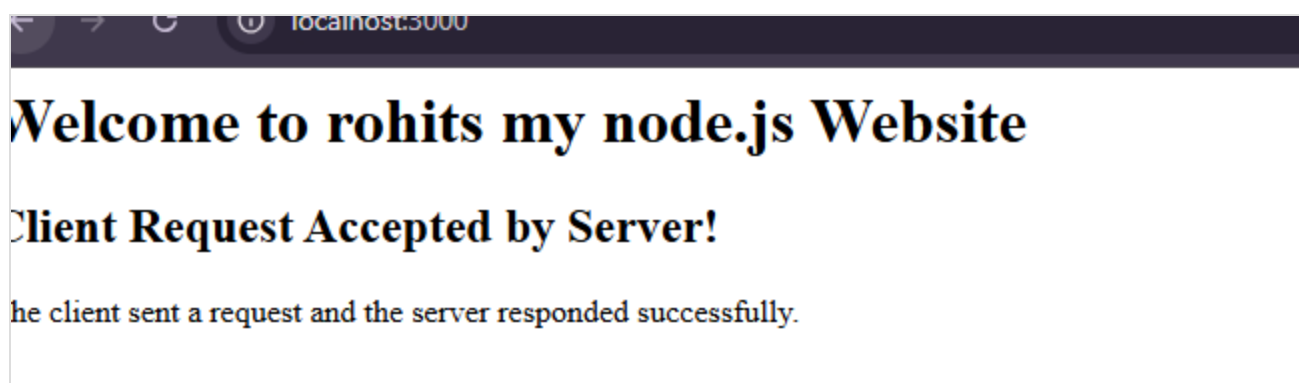
```
node.js v26.8.1

C:\Users\anits-csm\Desktop\NodeModule>node modules.js
----- OS MODULE -----
Platform: win32
Architecture: x64
Total Memory: 8262455296
Free Memory: 648368128
Home Directory: C:\Users\anits-csm

----- PATH MODULE -----
Directory: C:\Users\anits-csm\Desktop\67
File Name: me.text.txt
Extension: .txt
Joined Path: C:\Users\anits-csm\Desktop\NodeModule\data\file.txt

----- DNS MODULE -----

----- NET MODULE -----
Server running on port 3000
IP Address: 192.178.193.138
IP Family: 4
```



The screenshot shows a web browser window with the address bar set to 'localhost:3000'. The page content includes a heading 'Welcome to rohits my node.js Website', a subheading 'Client Request Accepted by Server!', and a paragraph 'The client sent a request and the server responded successfully.'.

localhost:3000

Welcome to rohits my node.js Website

Client Request Accepted by Server!

The client sent a request and the server responded successfully.

Result:

Program executed successfully.

Task 2

Aim: Node.js File Management: Create a Node.js application to write, read, append, and display file contents.

Source Code:

```
const fs = require('fs');
const rline = require('readline').createInterface({
  input: process.stdin,
  output: process.stdout
});

rline.question('Provide file name: ', (fName) => {
  rline.question('Provide file content: ', (fData) => {
    fs.writeFile(fName, fData, (error) => {
      if (error) return console.error(error);
      console.log('--- Data saved successfully ---');

      fs.readFile(fName, 'utf8', (error, readData) => {
        console.log('Current Data:', readData);

        rline.question('Provide data to append: ', (appendData) => {
          fs.appendFile(fName, '\n' + appendData, (error) => {
            fs.readFile(fName, 'utf8', (error, finalResult) => {
              console.log('Final Data:', finalResult);
              rline.close();
            });
          });
        });
      });
    });
  });
});
```

Output:

```

Node.js v26.8.1

C:\Users\anits-csm\Desktop\NodeModule>node modules.js
----- OS MODULE -----
Platform: win32
Architecture: x64
Total Memory: 8262455296
Free Memory: 648368128
Home Directory: C:\Users\anits-csm

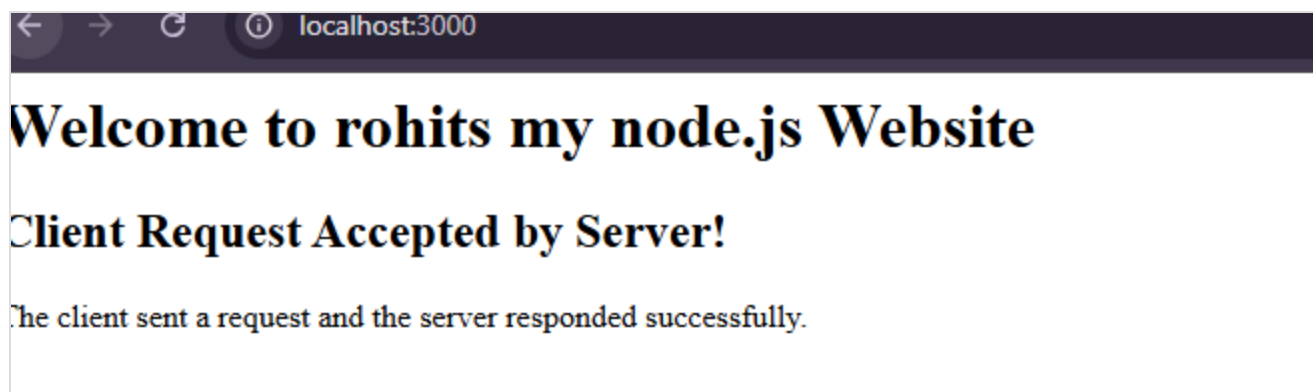
----- PATH MODULE -----
Directory: C:\Users\anits-csm\Desktop\67
File Name: me.text.txt
Extension: .txt
Joined Path: C:\Users\anits-csm\Desktop\NodeModule\data\file.txt

----- DNS MODULE -----

----- NET MODULE -----
Server running on port 3000
IP Address: 192.178.193.138
IP Family: 4

```

Test Cases:



```

C:\Users\anits-csm\Desktop\NodeModule>node server.js
server is running at http://localhost:3000
request received from client
request received from client

```

Result:

Program executed successfully.

Task 3

Aim: Express.js Student Server: Create routes /, /students, and /about to serve JSON and HTML.

Source Code:

```
const express = require('express');
const server = express();

const myClassmates = [
  { id: 101, name: 'Charishma', dept: 'CSE' },
  { id: 102, name: 'Rohit', dept: 'CSE' },
  { id: 103, name: 'Manoj', dept: 'CSE' },
  { id: 104, name: 'Somya', dept: 'CSE' },
  { id: 105, name: 'Kiran', dept: 'CSE' }
];

server.get('/', (req, res) => res.send('<h2>Student Portal Base URL</h2>'));
server.get('/students', (req, res) => res.json({ total: myClassmates.length, students: myClassmates }));
server.get('/about', (req, res) => res.send('<p>This route gives info about the server setup.</p>'));

server.listen(3000, () => console.log('Listening on port 3000...'));
```

Output:



Test Cases:

```
C:\Users\anils-csm\Desktop\nodemodule>node server.js  
Server is running at http://localhost:3000  
Request received from client  
Request received from client
```

```
C:\Users\rohit\OneDrive\Desktop\obs>node app.js
```

Result:

Program executed successfully.

Task 4

Aim: Document Object Model (DOM): Demonstrate dynamically changing content, style, and attributes.

Source Code:

```
<!DOCTYPE html>
<html>
<head>
  <style>
    .container { padding: 15px; border: 1px solid gray; }
    .highlight { background: yellow; border: 3px solid red; font-weight: bold; }
  </style>
</head>
<body>
  <div id="targetDiv" class="container">Original DOM Element</div>
  <button id="btnAction">Trigger DOM Manipulation</button>
  <script>
    document.getElementById('btnAction').onclick = function() {
      let tgt = document.getElementById('targetDiv');
      tgt.innerText = "DOM manipulation triggered successfully!";
      tgt.className = "container highlight";
      tgt.setAttribute("data-updated", "true");
    };
  </script>
</body>
</html>
```

Output:

```
C:\Users\anits-csm\Desktop\NodeModule>node server.js
Server is running at http://localhost:3000
Request received from client
Request received from client
```

Test Cases:

```
C:\Users\rohit\OneDrive\Desktop\obs>node app.js
```

```
// This logs to your VS Code terminal every time the page is refreshed
console.log('Request received from client');

// This sends the HTML content to the browser
res.writeHead(200, { 'Content-Type': 'text/html' });
res.write('<h1>Welcome to ALLIPIILLI ROHIT | my node.js Website</h1>');
res.write('<h2>Client Request Accepted by Server!</h2>');
res.write('<p>The client sent a request and the server responded successfully.</p>');
res.end();
});

server.listen(3000, () => {
  console.log('Server is running at http://localhost:3000');
});
```

Result:

Program executed successfully.